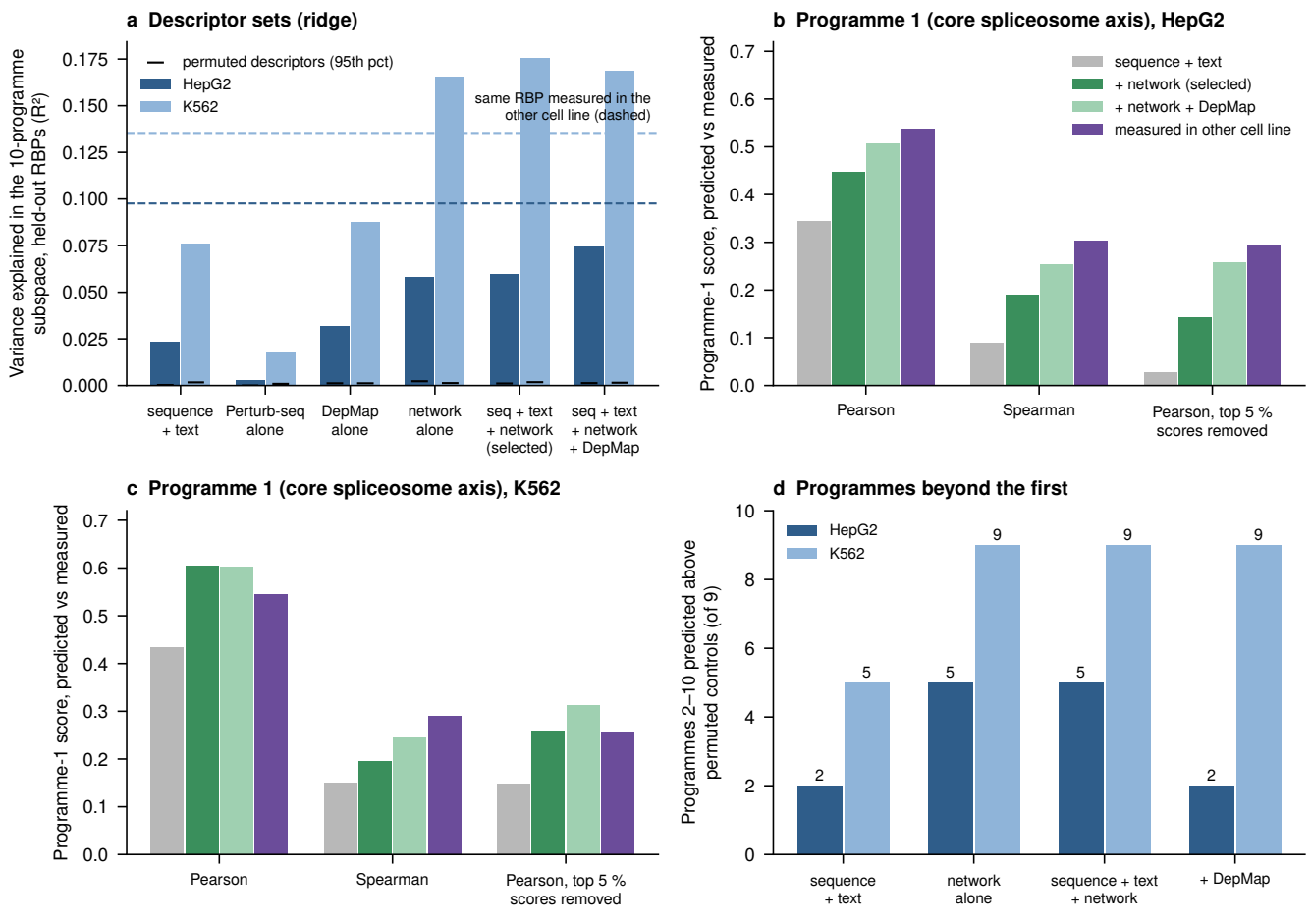
a Descriptor sets (ridge)
Variance explained in the 10-programme subspace, held-out RBPs (R²)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
permuted descriptors (95th pct)
HepG2
K562
same RBP measured in the other cell line (dashed)
sequence + text
Perturb-seq alone
DepMap alone
network alone
seq + text + network (selected)
seq + text + network + DepMap
b Programme 1 (core spliceosome axis), HepG2
Programme-1 score, predicted vs measured
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
sequence + text
+ network (selected)
+ network + DepMap
measured in other cell line
Pearson
Spearman
Pearson, top 5 % scores removed
c Programme 1 (core spliceosome axis), K562
Programme-1 score, predicted vs measured
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Pearson
Spearman
Pearson, top 5 % scores removed
d Programmes beyond the first
Programmes 2–10 predicted above permuted controls (of 9)
10
8
6
4
2
0
HepG2
K562
2
5
5
9
5
9
2
9
sequence + text
network alone
sequence + text + network
+ DepMap